

Curriculum Vitae for Benjamin Katz

CONTACT INFORMATION	E-mail: bkatz2@huksters.unl.edu Office: Avery Hall 337, University of Nebraska-Lincoln
EDUCATION	PhD Student, University of Nebraska, advised by Mark Walker, 2022-present M.S. in Mathematics, University of Nebraska, 2022-2024 B.S. in Mathematics, The University of Texas at Austin, 2018-2021
RESEARCH INTERESTS	I am interested in commutative algebra and homological algebra as well as their intersection with discrete math.
PAPERS	Cohomological support varieties of monomial ideals, with Kara Fagerstrom, Julianne Faur, Kesavan Mohana Sundaram, Stephen Stern, and Ryan Watson preprint: arXiv:2605.29103 Lifting systems for finite length modules, with Nawaj KC, Kesavan Mohana Sundaram, Andrew J. Soto Levins, and Ryan Watson submitted, preprint: arXiv:2602.01440 On the Ext Analog of the Euler Characteristic, with Andrew Soto Levins Journal of Pure and Applied Algebra, January 2026
INVITED TALKS	AMS Sectional, University of Kansas, Special Session on Algebraic and Homological Perspectives with a view towards Geometry III, Spring 2025
SEMINAR TALKS	Commutative Algebra Seminar, University of Nebraska-Lincoln, Spring 2025 Commutative Algebra Reading Seminar, University of Nebraska-Lincoln, Fall 2024 Commutative Algebra Reading Seminar, University of Nebraska-Lincoln, Spring 2024
SERVICE	Co-organizer for UNL CARS, Commutative Algebra Reading Seminar, Fall 2025-Spring 2026 Co-organizer for URiCA conference, Upcoming Researchers in Commutative Algebra, Fall 2024-present UNL Math Directed Reading Program mentor, Fall 2023
CONFERENCES ATTENDED	A Celebration of Golod's Mathematical Legacy, University of Warwick, Spring 2026 CA+, Iowa State University, Spring 2026 KUMUNU, University of Missouri, Fall 2025, CA+, Iowa State University, Spring 2025 KUMUNU, University of Missouri, Fall 2024 Math 125, University of Nebraska-Lincoln, Spring 2023
TEACHING EXPERIENCE	At University of Nebraska-Lincoln Spring 2025, Math 221-Differential Equations, Recitation Instructor Fall 2024, Math 103-College Algebra and Trigonometry, Instructor of Record Spring 2024, Math 203-Contemporary Mathematics, Instructor of Record Fall 2023, Math 102-Trigonometry, Instructor of Record Spring 2023, Math 106-Calculus 1, Recitation Instructor Fall 2022, Math 106-Calculus 1, Recitation Instructor